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Paper 1: Optimizing Subcutaneous Antibody Dosing Regimens Through Operating Space Maps: r

Bibliographic Information

- **Title:** Optimizing Subcutaneous Antibody Dosing Regimens Through Operating Space Maps: rHuPH20 Case Study
- **Authors:** Nolan RP, Chintakuntla H
- **Journal:** Journal of Pharmacokinetics and Pharmacodynamics, 2026, 53:33
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Quick Take

This paper introduces Operating Space Maps, a visual PK simulation framework that translates complex pharmacokinetic data into an intuitive dose–frequency landscape for subcutaneous antibody delivery. Using rHuPH20 as a case study, the maps show how to convert IV regimens to SC while maintaining exposure and reducing C_{max}, and how to extend SC dosing intervals. This is essential reading for pharmacometricians, clinical pharmacologists, and drug development teams involved in early-stage formulation and regimen decisions for biologics.

Executive Summary

This paper introduces Operating Space Maps, a visual PK simulation framework that organizes multiple dosing scenarios into an intuitive landscape of dose–frequency options for subcutaneous (SC) antibody delivery. Using recombinant human hyaluronidase PH20 (rHuPH20) as a case study, the authors apply a two-compartment PK model with first-order SC absorption to simulate representative antibody regimens. The maps enable rapid assessment of IV-to-SC conversion and SC dose-interval extension, making trade-offs between C_{min}, C_{avg}, and C_{max} explicit while integrating practical SC delivery constraints such as injection volume. The methodology is broadly applicable to other SC-enabling technologies and is designed to guide early strategic decisions by cross-functional development teams.

Scientific Context & Motivation

Therapeutic antibody dosing is often constrained by formulation and administration factors, particularly for subcutaneous delivery where injection volume limits dose size. While rHuPH20 enables high-volume SC administration, development teams lack tools to quantify the strategic impact of such technologies. Traditional PK simulations are typically presented in formats that are difficult for non-modelers to interpret and do not integrate practical delivery constraints (e.g., device volume limits). This work addresses the gap by providing a visual framework that makes

trade-offs between dose, frequency, and delivery feasibility explicit, enabling cross-functional teams to evaluate regimen options early in development.

✂ Methodological Snapshot

The authors employ a two-compartment PK model with first-order SC absorption, parameterized using values from a prior meta-analysis of ten clinical antibodies. For rHuPH20-enabled SC delivery, the absorption rate constant (k_a) and bioavailability (F) are adjusted based on that same analysis. The framework generates Operating Space Maps by simulating a benchmark regimen (e.g., 1000 mg IV) at multiple schedules (QW to Q24W) and then identifying SC doses with rHuPH20 that match the benchmark’s C_{avg} at each schedule. The resulting grid of PK profiles is color-coded by injection volume (assuming 150 mg/mL formulation), enabling rapid visual assessment of practical delivery feasibility. The methodology is deterministic (no IIV) and uses generic parameters, making it suitable for early strategic triage rather than late-stage regulatory submissions.

Detailed Methodological Analysis

Modeling Approach

A two-compartment PK model with first-order SC absorption was used. The model includes SC depot (A_0), central (C_1), and peripheral (C_2) compartments. Parameters: $V_1 = 3.2$ L, $V_2 = 2.8$ L, $CL = 0.21$ L/day, $Q = 0.48$ L/day, $k_a = 0.29$ day⁻¹ (IV), 0.46 day⁻¹ (SC with rHuPH20), $F = 0.70$ (SC without rHuPH20), 0.80 (SC with rHuPH20). Simulations were performed in MATLAB.

Data Sources

Dose regimen landscape data were extracted from approved product labels via DailyMed and PharmaCircle using GPT-4o mini LLM with five independent extractions per label; only regimens appearing in $\geq 4/5$ extractions were retained. The database included 669 products and 1704 unique dosing regimens. For PK simulations, parameter values were derived from a previously published PopPK analysis of ten clinical antibodies (Nolan & Printz, 2024).

Estimation Methods

No estimation was performed; this is a simulation study. The two-compartment ODE model was solved using MATLAB’s ode15s with default tolerances, integrating segment-by-segment between dose times on a daily output grid.

Model Evaluation

External validation was performed by comparing simulated trends to the PK of a recently approved rHuPH20-coformulated product, SC ocrelizumab (Ocrevus Zunovo). The label-supported trends—comparable C_{avg} , reduced C_{max} , and feasible high-volume SC delivery—aligned with the simulation outputs. No formal internal validation (e.g., VPC, bootstrap) was conducted as this is a deterministic simulation framework.

Covariate Analysis

No covariate analysis was performed. The simulations used fixed, representative mean parameters without inter-individual variability. The authors note that as population variability and covariance structures become available for a given asset, the same workflow can incorporate IIV via Monte Carlo simulation.

Statistical Rigor Assessment

This is a deterministic simulation study, not a statistical analysis. No formal hypothesis testing, confidence intervals, or uncertainty quantification are provided. The use of mean parameters without IIV means the profiles represent population averages, not predictions for individual patients. The external validation is qualitative, comparing simulated trends to a single approved product (ocrelizumab). The LLM-based data extraction for the landscape analysis employed a reasonable quality control (5 independent extractions, $\geq 4/5$ consistency threshold), but the potential for residual errors or biases in the dose regimen database is not quantified. For its intended purpose (early strategic triage), the lack of statistical rigor is acceptable, but users should be aware that these maps do not replace formal PopPK analyses for regulatory decision-making.

Key Findings

The paper presents Operating Space Maps as a novel visual framework for evaluating SC antibody dosing regimens. Key findings include: (1) For IV-to-SC conversion, a 1400 mg SC dose with rHuPH20 matches the C_{avg} of a 1000 mg IV Q4W regimen while reducing C_{max} by approximately 50%, offering potential safety advantages. (2) SC dose-interval extension is feasible with rHuPH20: a 300 mg SC Q4W regimen can be extended to Q8W with a 540 mg SC dose, though with wider C_{max} - C_{min} fluctuations that must be evaluated against the therapeutic window. (3) The current landscape analysis (Figure 3) shows that 75% of SC antibody doses are ≤ 300 mg, reflecting volume constraints of conventional SC delivery, while rHuPH20 enables doses up to 1875 mg in volumes up to 23 mL. (4) External validation using ocrelizumab (Ocrevus Zunovo) confirms the qualitative trends: comparable AUC, reduced C_{max} , and feasible high-volume SC delivery. (5) The maps make explicit the trade-offs between dose, frequency, and delivery feasibility, enabling rapid identification of regimens that are both pharmacokinetically suitable and practically deliverable.

Clinical & Regulatory Implications

The maps directly inform TPP optimization by making trade-offs between C_{min} , C_{avg} , and C_{max} explicit. For IV-to-SC conversion, they show that a 1400 mg SC dose with rHuPH20 can match the C_{avg} of a 1000 mg IV Q4W regimen while reducing C_{max} by $\sim 50\%$, potentially mitigating C_{max} -driven toxicities. For SC interval extension, a 300 mg SC Q4W regimen can be extended to Q8W with a 540 mg SC dose using rHuPH20, though the wider peak-to-trough swings must be evaluated against the therapeutic window. The framework also highlights that SC doses exceeding ~ 300 mg (at 150 mg/mL) require volumes > 2 mL, necessitating rHuPH20 for rapid injection.

Regulators may find these maps useful for contextualizing proposed SC regimens during early-stage interactions.

Strengths & Limitations

Strengths

- Novel visual framework that translates complex PK simulations into an intuitive, decision-ready format accessible to non-modelers.
- Explicit integration of practical SC delivery constraints (injection volume, device limits) with PK metrics.
- Systematic evaluation of a wide range of dose-frequency combinations (QW to Q24W) in a single, navigable map.
- Methodology is technology-agnostic and can be applied to other SC-enabling approaches beyond rHuPH20.
- External validation using a recently approved rHuPH20-coformulated product (ocrelizumab) supports the qualitative trends.
- Clear workflow (Table 2) enables readers to implement the framework for their own assets.

Limitations (Acknowledged by Authors)

- Simulations use generic, representative parameters derived from prior analyses, not asset-specific data.
- Deterministic profiles without inter-individual variability (IIV); prediction intervals are not provided.
- The framework does not incorporate mechanistic details such as target-mediated drug disposition (TMDD) or physiologically-based PK (PBPK).
- External validation is limited to a single product (ocrelizumab).
- The dose regimen landscape dataset is derived from licensed data (PharmaCircle) and is not publicly available.

Limitations (Expert Review)

- The assumption of PK linearity across all dose levels may not hold for antibodies with nonlinear clearance due to target binding.
- The use of a fixed 150 mg/mL formulation concentration may not reflect the range of achievable concentrations for all antibodies.
- The framework does not account for immunogenicity, which can alter PK over time, particularly for SC administration.
- The maps do not consider patient-specific factors (e.g., body weight, renal function) that may influence exposure.
- The LLM-based data extraction, while quality-controlled, may still contain errors or miss nuances in complex dosing regimens (e.g., weight-based dosing, loading doses).
- The simulation assumes instantaneous absorption for SC doses, which may not capture the full time course of absorption for very high volumes.

Generalizability

The framework is broadly applicable to any SC-enabling technology (e.g., high-concentration formulations, half-life extension) for which foundational PK data exist. However, the specific numerical results (dose matching, volume requirements) depend on the generic parameters used, which may not apply to antibodies with nonlinear PK (e.g., TMDD) or atypical absorption characteristics. The methodology is best suited for early-stage decisions and should be complemented by asset-specific modeling for later-stage programs.

Key Equations

SC Depot Compartment

$$\frac{dA_0}{dt} = -k_a \cdot A_0$$

Differential equation for the SC depot compartment (A_0), representing the amount of antibody at the injection site.

Central Compartment

$$\frac{dC_1}{dt} = \frac{k_a \cdot A_0}{V_1} - k_{el} \cdot C_1 - k_{12} \cdot C_1 + k_{21} \cdot C_2 \cdot \frac{V_2}{V_1}$$

Differential equation for the central compartment concentration (C_1), accounting for absorption from depot, elimination, and distribution to the peripheral compartment.

Peripheral Compartment

$$\frac{dC_2}{dt} = k_{12} \cdot C_1 \cdot \frac{V_1}{V_2} - k_{21} \cdot C_2$$

Differential equation for the peripheral compartment concentration (C_2), representing distribution from and back to the central compartment.

Elimination Rate Constant

$$k_{el} = \frac{CL}{V_1}$$

Elimination rate constant derived from clearance (CL) and central volume (V_1).

Distribution Rate Constants

$$k_{12} = \frac{Q}{V_1}, k_{21} = \frac{Q}{V_2}$$

Inter-compartmental distribution rate constants derived from inter-compartmental clearance (Q) and compartment volumes.

SC Dose Conversion to nmol

$$\text{Dose}_{\text{nmol}} = \text{Dose}_{\text{mg}} \times \frac{10^6}{\text{MW}} \times F$$

Conversion of SC dose (mg) to nmol for simulation, accounting for bioavailability (F) and molecular weight ($\text{MW} = 150,000 \text{ g/mol}$).

IV Dose Conversion to nM

$$C_{0, \text{IV}} = \frac{\text{Dose}_{\text{mg}} \times 10^6 / \text{MW}}{V_1}$$

Conversion of IV dose (mg) to nM concentration by dividing by central volume V_1 .

Figures & Tables

- **Figure 1:** Schematic of rHuPH20 mechanism: depolymerization of hyaluronan in the SC space enables rapid administration of high-volume therapeutics.
 - *Significance:* Provides the biological rationale for rHuPH20's ability to overcome SC volume constraints.
- **Figure 2:** Two-compartment PK model diagram with SC depot (A_0), central (C_1), and peripheral (C_2) compartments, showing absorption (k_a), elimination (k_{el}), and distribution (k_{12} , k_{21}) rate constants.
 - *Significance:* Defines the structural model used for all simulations; essential for understanding the PK framework.
- **Figure 3:** Scatter plot of dose vs. dosing interval for IV (red) and SC (blue) antibody therapeutics from clinical development through marketed products.
 - *Significance:* Establishes the current landscape of antibody dosing regimens, highlighting the volume constraints on SC delivery (most SC doses $\leq 300 \text{ mg}$).
- **Figure 4:** Operating Space Map for converting a 1000 mg IV benchmark regimen to SC with rHuPH20. Each cell shows PK profiles for a given schedule (rows: IV benchmark; columns: SC with rHuPH20), with background color indicating injection volume.
 - *Significance:* Central figure demonstrating the framework's utility for IV-to-SC conversion; shows that 1400 mg SC Q4W matches 1000 mg IV C_{avg} with $\sim 50\%$ lower C_{max} .
- **Figure 5:** Operating Space Map for extending a 300 mg SC (without rHuPH20) benchmark regimen to SC with rHuPH20 at extended intervals.
 - *Significance:* Illustrates SC interval extension; shows that 300 mg SC Q4W can be extended to Q8W with 540 mg SC using rHuPH20, but with wider C_{max} - C_{min} swings.
- **Figure 6:** Operating Space Map for a 1000 mg SC (without rHuPH20) benchmark, showing expanded options with rHuPH20.
 - *Significance:* Demonstrates that even at higher SC doses, rHuPH20 enables further interval extension and volume reduction.
- **Table 1:** Parameter values for the two-compartment PK model: $V_1 = 3.2 \text{ L}$, $V_2 = 2.8 \text{ L}$, $CL = 0.21 \text{ L/day}$, $Q = 0.48 \text{ L/day}$, $k_a = 0.29 \text{ day}^{-1}$ (IV), 0.46 day^{-1} (SC with rHuPH20), $F = 0.70$ (SC without rHuPH20), 0.80 (SC with rHuPH20).

- *Significance*: Provides the numerical values used in all simulations; critical for reproducibility and understanding the assumed PK behavior.
 - **Table 2**: Workflow for creating an Operating Space Map: define benchmark regimen, simulate benchmark PK at multiple schedules, simulate C_{avg} -matched SC with rHuPH20 regimens, color-code by injection volume.
 - *Significance*: Methodological roadmap that enables readers to implement the framework for their own assets.
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Code & Reproducibility Assessment

The authors state that the dose regimen landscape dataset is derived from DailyMed (public) and PharmaCircle (licensed) and is not available from the authors. Supplementary Code S1 is referenced but not provided in the extracted content. MATLAB code for simulations is mentioned but not publicly deposited. Reproducibility is limited without access to the licensed data and the full simulation code.

Supplementary Materials

Supplementary material includes Supplementary Tables S1–S3 (noncompartmental metrics for all schedules in Figures 4–6) and Supplementary Code S1 (MATLAB simulation code). These are referenced in the main text but were not fully extracted in the provided content. The supplementary PDF (407 KB) is available for download from the journal website.

Future Directions

The authors suggest several extensions: (1) incorporating inter-individual variability (IIV) via Monte Carlo simulation to produce prediction intervals around $C_{max,ss}$, $C_{avg,ss}$, and $C_{min,ss}$; (2) integrating mechanistic details such as target-mediated drug disposition (TMDD) or physiologically-based PK (PBPK) models; (3) applying the framework to other SC-enabling technologies (e.g., high-concentration formulations, half-life extension); (4) matching alternative PK/PD drivers such as target occupancy rather than C_{avg} or C_{min} ; and (5) calibrating and validating predictions with emerging clinical data. Future studies could also explore the impact of different antibody concentrations (e.g., 100 mg/mL vs. 200 mg/mL) on the feasible operating space and extend the maps to include combination therapies.

Expert Commentary

This work fills a genuine gap in early-stage biopharmaceutical development: translating complex PK simulations into actionable, cross-functional decisions. The Operating Space Map concept is elegant in its simplicity—overlaying PK profiles with practical delivery constraints (volume, device limits) in a grid format that non-modelers can immediately grasp. From a senior pharmacometrics perspective, the deterministic approach (no IIV) is appropriate for the stated purpose of early strategic triage, but teams must remember that these are population-mean profiles, not predictions for individual patients. The reliance on generic parameters from a prior meta-analysis

is a reasonable starting point, but asset-specific TMDD or PBPK models will be essential for later-stage decisions, especially for antibodies with nonlinear PK. The external validation using ocrelizumab (Ocrevus Zunovo) is reassuring but limited to a single example; more extensive validation across diverse antibodies would strengthen confidence. The framework’s true value lies in its ability to foster dialogue between modelers, clinicians, and commercial teams, potentially reducing the number of scenarios that need full PopPK simulation. I would encourage users to extend the maps to include PD endpoints (e.g., target occupancy) and to incorporate uncertainty intervals as soon as IIV data become available.

Bottom Line

Operating Space Maps provide a visually intuitive, decision-ready framework that translates PK simulations into a dose–frequency landscape, explicitly incorporating SC delivery constraints (volume, device limits). For rHuPH20-enabled SC regimens, the maps demonstrate that equivalent average exposure to IV benchmarks can be achieved with reduced C_{max} and, in many cases, extended dosing intervals. This tool enables cross-functional teams to rapidly triage feasible SC regimens early in development, reserving detailed asset-specific PopPK/TMDD modeling for only the most promising candidates.

☒ Figures

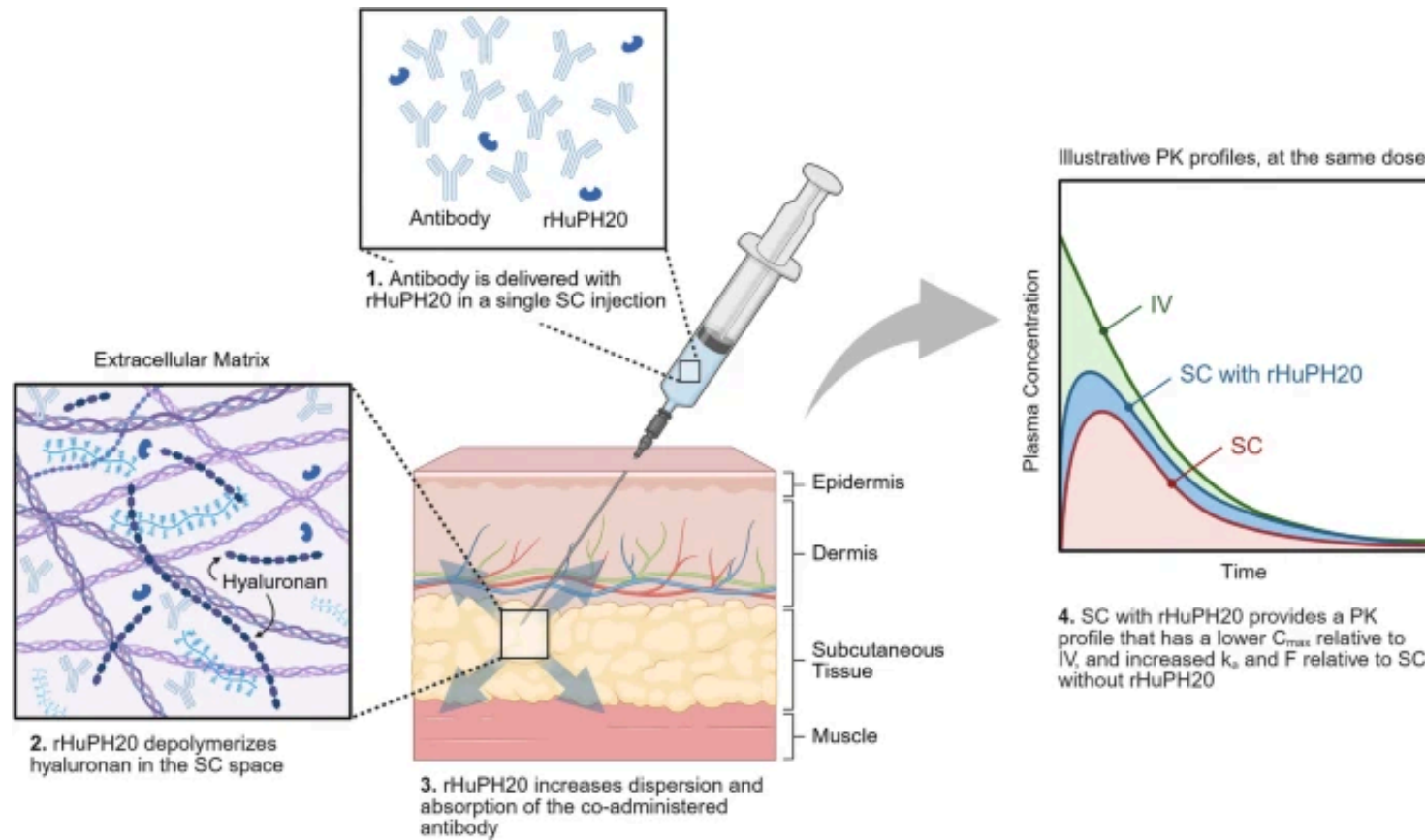
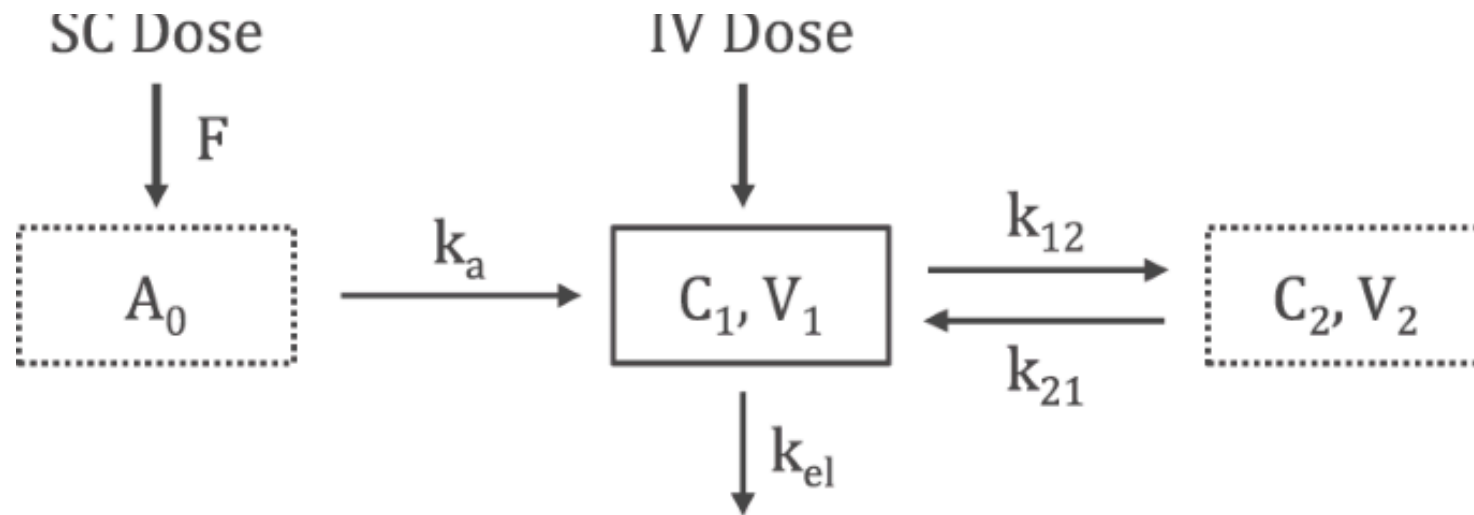


Figure 1: Figure 1



$$\frac{dA_0}{dt} = -k_a A_0$$

$$\frac{dC_1}{dt} = \frac{k_a A_0}{V_1} - k_{12} C_1 + k_{21} C_2 \frac{V_2}{V_1} - k_{el} C_1$$

$$\frac{dC_2}{dt} = k_{12} C_1 \frac{V_1}{V_2} - k_{21} C_2$$

Figure 2: Figure 2

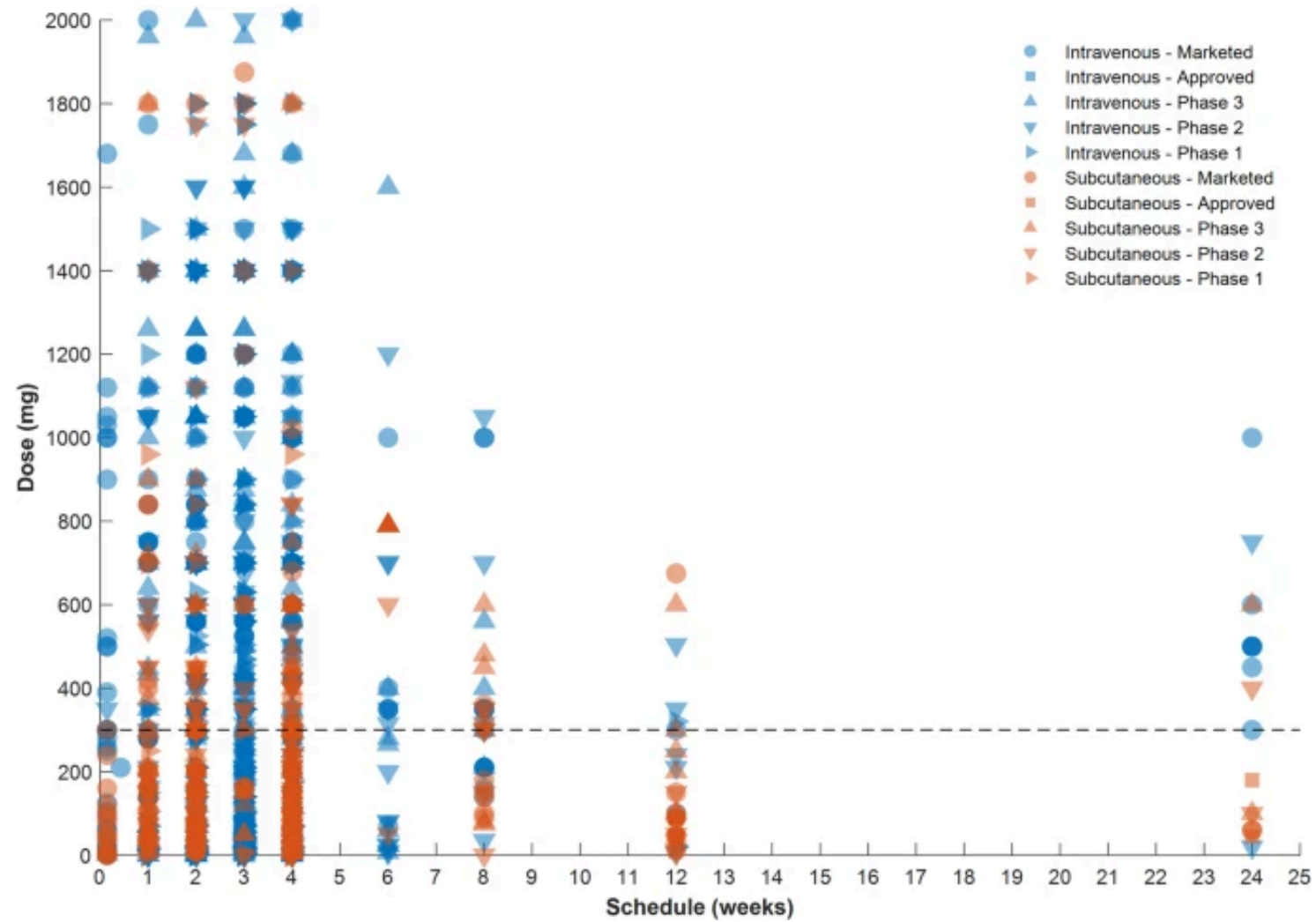


Figure 3: Figure 3

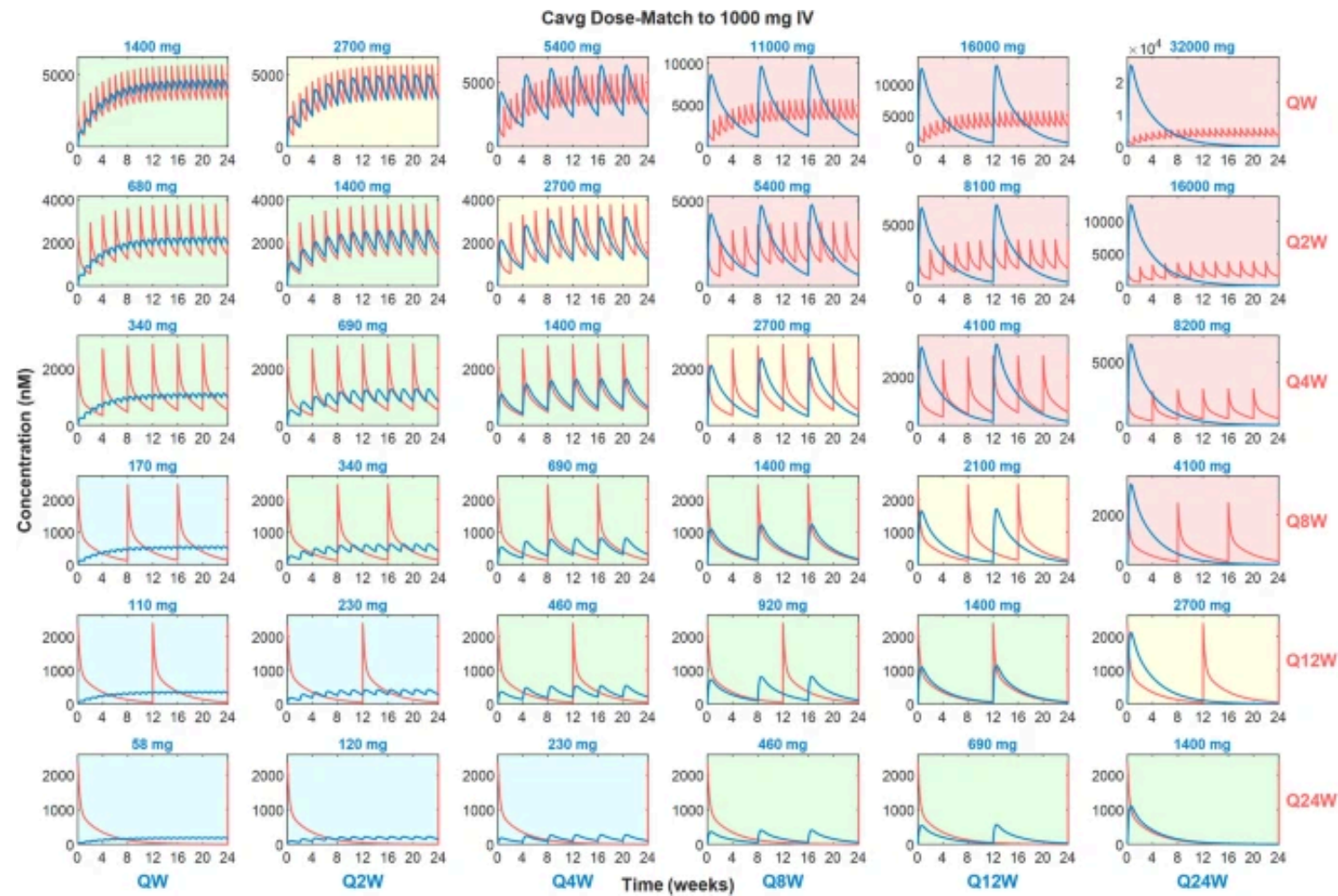


Figure 4: Figure 4

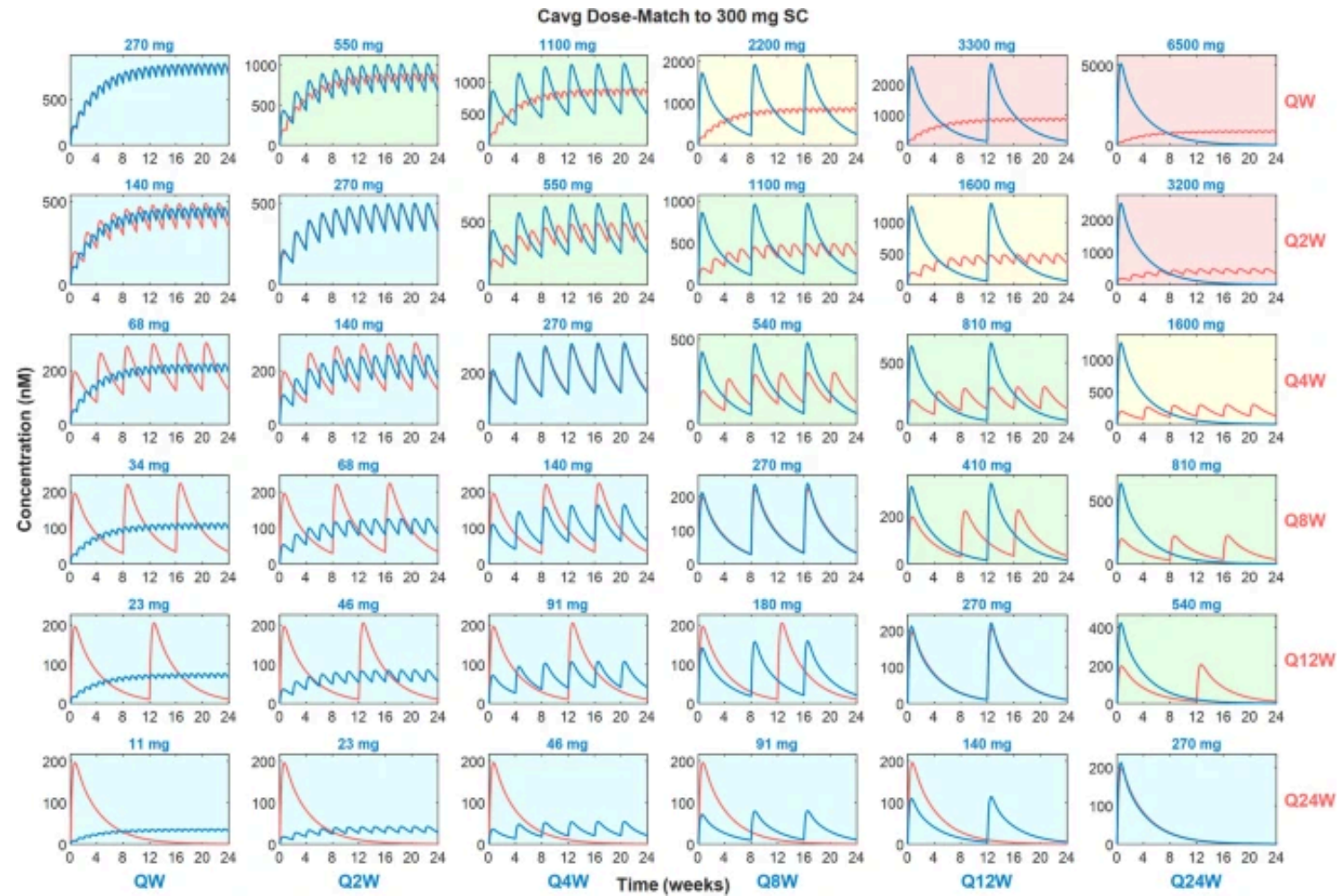


Figure 5: Figure 5

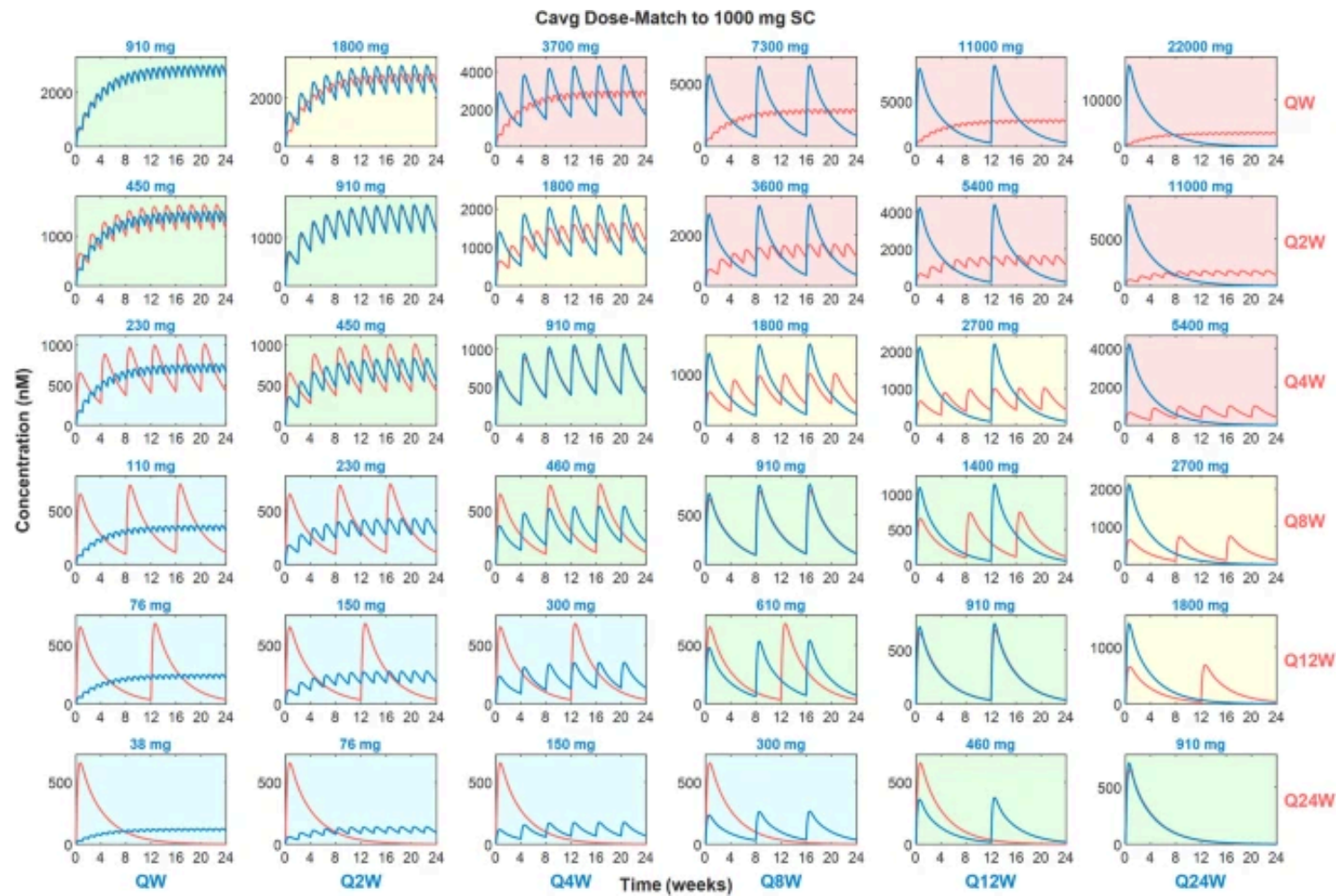


Figure 6: Figure 6